PROGRAMMING ASSIGNMENT 1

Subject: Algorithm Complexity Analysis

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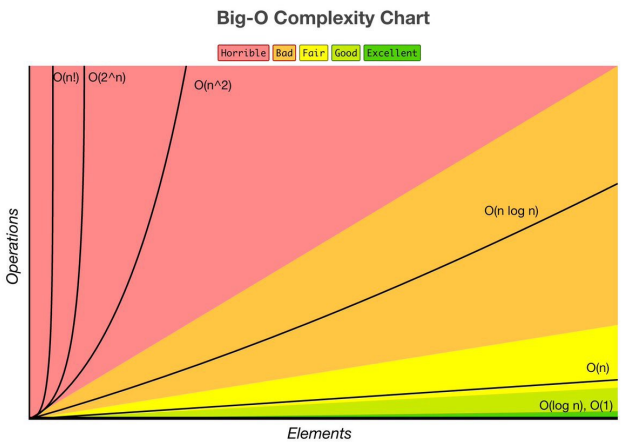
Programming Language: Java (OpenJDK 11)
Due Date: **Friday, 20.03.2026 (23:59:59)**



1 Introduction

Analysis of algorithms is the area of computer science that provides tools to analyze the efficiency of different methods of solutions. The efficiency of an algorithm is primarily evaluated in terms of time complexity and space complexity. Analysis of algorithms is mainly used to predict performance and compare algorithms that are developed for the same task. Also it provides guarantees for performance and helps to understand theoretical basis.

A complete analysis of the running time of an algorithm involves the following steps:

- Implement the algorithm completely.
- Determine the time required for each basic operation.
- Identify unknown quantities that can be used to describe the frequency of execution of the basic operations.
- Develop a realistic model for the input to the program.
- Analyze the unknown quantities, assuming the modeled input.

- Calculate the total running time by multiplying the time by the frequency for each operation, then adding all the products.

> In this experiment, you will analyze different sorting algorithms and compare their running times on a number of inputs with changing sizes.

2 Background and Problem Definition

Efficient sorting is important for optimizing the efficiency of other algorithms (such as search and merge algorithms) that require input data to be sorted. The efficiency of a sorting algorithm can be observed by applying it to sort datasets of varying sizes and other characteristics of the dataset instances that are to be sorted. In this assignment, you will be classifying the given sorting algorithms based on two criteria:

- **Computational (Time) Complexity**: Determining the best, worst and average case behavior in terms of the size of the dataset. Table 1 illustrates a comparison of computational complexity of some well-known sorting algorithms.

Table 1: Computational complexity comparison of some well-known algorithms.

Algorithm	**Best Case**	**Average Case**	**Worst Case**
Selection Sort	$\Omega(n^2)$	$\Theta(n^2)$	$O(n^2)$
Bubble Sort	$\Omega(n)$	$\Theta(n^2)$	$O(n^2)$
Heap Sort	$\Omega(n \log n)$	$\Theta(n \log n)$	$O(n \log n)$
Comb Sort	$\Omega(n \log n)$	$\Theta(n^2)$	$O(n^2)$

- **Auxiliary Memory (Space) Complexity**: Some sorting algorithms are performed "in-place" using swapping. An in-place sort needs only $O(1)$ auxiliary memory apart from the memory used for the items being sorted. On the other hand, some algorithms may need $O(\log n)$ or $O(n)$ auxiliary memory for sorting operations. Table 2 illustrates an auxiliary space complexity comparison of the same well-known sorting algorithms.

Table 2: Auxiliary space complexity comparison of some well-known algorithms for the average case.

Algorithm	**Auxiliary Space Complexity**
Selection Sort	$O(1)$
Bubble Sort	$O(1)$
Heap Sort	$O(1)$
Comb Sort	$O(1)$

A time complexity analysis focuses on gross differences in the efficiency of algorithms that are likely to dominate the overall cost of a solution. See the example given below:

Code	Unit Cost	Times
i=1;	c1	1
sum = 0;	c2	1
while (i $\leq$ n) {	c3	$n+1$
j=1;	c4	n
while (j $\leq$ n) {	c5	$n \cdot (n+1)$
sum = sum + i;	c6	$n \cdot n$
j = j + 1;	c7	$n \cdot n$
}		
i = i +1;	c8	n
}		

The total cost of the given algorithm is $c1 + c2 + (n+1) \cdot c3 + n \cdot c4 + n \cdot (n+1) \cdot c5 + n \cdot n \cdot c6 + n \cdot n \cdot c7 + n \cdot c8$. The running time required for this algorithm is proportional to n^2, which is determined as its growth rate, and it is usually denoted as $O(n^2)$.

3 Assignment Tasks

The main objective of this assignment is to show the relationship between the running time of the algorithm implementations with their theoretical asymptotic complexities. You are expected to implement the sorting algorithms given as pseudocodes, and perform a set of experiments on the given datasets to show that the empirical data follows the corresponding asymptotic growth functions. To do so, you will have to consider how to reduce the noise in your running time measurements and plot the results to demonstrate and analyze the asymptotic complexities.

3.1 Sorting Algorithms to Implement

- **Comparison-based sorting algorithms**: In comparison-based sorting algorithms, the elements are compared to determine their order in the final sorted output. All comparison-based sorting algorithms have a complexity lower bound of $\Omega(n \log n)$. You will implement the following comparison-based sorting algorithms:
  - Quick sort (Iterative Implementation) given in Alg. 1
  - Insertion sort given in Alg. 2
  - Merge sort (Iterative Implementation) given in Alg. 3
  - Shell sort given in Alg. 4
- **Non-comparison-based sorting algorithms**: Some algorithms sort data without direct element comparisons, often leveraging structure or predefined assumptions. Non-comparison-based sorting algorithms may perform in $O(n)$ time complexity. You will implement the following non-comparison-based sorting algorithm:
  - Radix sort given in Alg. 5

Algorithm 1 Quick Sort (Iterative Implementation)

```
procedure QUICK-SORT(A: array, low: integer, high: integer)
    stackSize ← high - low + 1
    stack ← new array of size stackSize
    top ← -1
    top ← top + 1
    stack[top] ← low
    top ← top + 1
    stack[top] ← high
    while top ≥ 0 do
        high ← stack[top]
        top ← top - 1
        low ← stack[top]
        top ← top - 1
        pivot ← PARTITION(A, low, high)
        if pivot - 1 > low then
            top ← top + 1
            stack[top] ← low
            top ← top + 1
            stack[top] ← pivot - 1
        end if
        if pivot + 1 < high then
            top ← top + 1
            stack[top] ← pivot + 1
            top ← top + 1
            stack[top] ← high
        end if
    end while
end procedure

procedure PARTITION(A: array, low: integer, high: integer)
    pivot ← A[high]
    i ← low - 1
    for j ← low to high - 1 do
        if A[j] ≤ pivot then
            i ← i + 1
            swap(A[i], A[j])
        end if
    end for
    swap(A[i+1], A[high])
    return i + 1
end procedure
```

Algorithm 2 Insertion Sort

```
procedure INSERTION-SORT(A: array)
    for j← 1, . . . , length(A) do
        key ← A[j]
        i ← j - 1
        while i >= 0 and A[i] > key do
            A[i + 1] ← A[i]
            i ← i - 1
        end while
        A[i + 1] ← key
    end for
end procedure
```

Algorithm 3 Merge Sort (Iterative Implementation)

```
procedure MERGE-SORT(A: array)
    n ← length(A)
    temp ← new array of size n
    currSize ← 1
    while currSize < n do
        leftStart ← 0
        while leftStart < n - 1 do
            mid ← min(leftStart + currSize - 1, n - 1)
            rightEnd ← min(leftStart + 2 * currSize - 1, n - 1)
            MERGE(A, temp, leftStart, mid, rightEnd)
            leftStart ← leftStart + 2 * currSize
        end while
        currSize ← 2 * currSize
    end while
end procedure

procedure MERGE(A: array, temp: array, left: integer, mid: integer, right: integer)
    i ← left
    j ← mid + 1
    k ← left
    while i ≤ mid and j ≤ right do
        if A[i] ≤ A[j] then
            temp[k] ← A[i]
            i ← i + 1
        else
            temp[k] ← A[j]
            j ← j + 1
        end if
        k ← k + 1
    end while
    while i ≤ mid do
        temp[k] ← A[i]
        i ← i + 1
        k ← k + 1
    end while
    while j ≤ right do
        temp[k] ← A[j]
        j ← j + 1
        k ← k + 1
    end while
    for i ← left to right do
        A[i] ← temp[i]
    end for
end procedure
```

Algorithm 4 Shell Sort

```
procedure SHELLSORT(A: array)
    n ← length(A)
    h ← 1
    while h < n / 3 do
        h ← 3h+1
    end while
    while h <= 1 do
        for  i ← h to n-1 do
            for  j ← i to -h , j >= h & less( A[j], A[j-h]) do
                exch(A, j, j-h)
            end for
        end for
    end while
end procedure
```

Algorithm 5 Radix Sort

```
procedure RADIXSORT(A: array, d: number of digits)
    for pos ← 1 to d do                                  ▷ Process each digit position
        A ← COUNTINGSORT(A, pos)
    end for
    return A
end procedure
procedure COUNTINGSORT(A: array, pos: digit position)
    count ← array of 10 zeros                            ▷ Assuming decimal digits (0-9)
    output ← array of the same length as A
    size ← length(A)
    for i ← 1, . . . , size do
        digit ← getDigit(A[i], pos)
        count[digit] ← count[digit] + 1
    end for
    for i ← 2, . . . , 10 do
        count[i] ← count[i] + count[i − 1]
    end for
    for i ← size, . . . , 1 do
        digit ← getDigit(A[i], pos)
        count[digit] ← count[digit] −1
        output[count[digit]] ← A[i]
    end for
    return output
end procedure
```

3.2 Dataset

You will test the given sorting algorithms on a real-world financial dataset containing 5 years of historical stock market data. This dataset (all_stocks_5yr.csv) includes daily trading records for multiple companies over a 5-year period, with approximately 850,000 records of stock price information. This dataset captures the daily opening, closing, high, low prices and trading volumes for major stocks like Apple (AAPL), Microsoft (MSFT), Google (GOOG), Amazon (AMZN), and many others from leading technology companies.

In order to be able to perform a comparative analysis of the performance of the given sorting algorithms over different data sizes, you will consider several smaller partitions of the dataset, that is, its subsets of sizes 500, 1000, 2000, 4000, 8000, 16000, 32000, 64000, 128000, and 250000 starting from the beginning of the file. You will be sorting the records based on the **volume** feature given in the 6th column (**volume**), which is of type *int* (see Fig. 1).

date	open	high	low	close	volume	Name
2013-02-08	15.07	15.12	14.63	14.75	8408	AAL
2013-02-11	14.89	15.01	14.26	14.46	8882	AAL
2013-02-12	14.45	14.51	14.1	14.27	8126	AAL
2013-02-13	14.3	14.94	14.25	14.66	10260	AAL
2013-02-14	14.94	14.96	13.16	13.99	31880	AAL
2013-02-15	13.93	14.61	13.93	14.5	15628	AAL
2013-02-19	14.33	14.56	14.08	14.26	11354	AAL
2013-02-20	14.17	14.26	13.15	13.33	14725	AAL
2013-02-21	13.62	13.95	12.9	13.37	11922	AAL
2013-02-22	13.57	13.6	13.21	13.57	6071	AAL
2013-02-25	13.6	13.76	13	13.02	7186	AAL
2013-02-26	13.14	13.42	12.7	13.26	9419	AAL
2013-02-27	13.28	13.62	13.18	13.41	7390	AAL
2013-02-28	13.49	13.63	13.39	13.43	6144	AAL
2013-03-01	13.37	13.95	13.32	13.61	7377	AAL
2013-03-04	13.5	14.07	13.47	13.9	8175	AAL
2013-03-05	14.01	14.05	13.71	14.05	7676	AAL
2013-03-06	14.52	14.68	14.25	14.57	13243	AAL
2013-03-07	14.7	14.93	14.5	14.82	9125	AAL
2013-03-08	14.99	15.2	14.84	14.92	10594	AAL
2013-03-11	14.85	15.15	14.71	15.13	6962	AAL
2013-03-12	15.14	15.6	14.95	15.5	8999	AAL
2013-03-13	15.54	16.2	15.48	15.91	11380	AAL

Figure 1: Sorting the data in the volume column only.

3.3 Experiments and Analysis Tasks

Assignment Steps Summarized:

- Implement the given algorithms in Java.
- Perform the experiments on the given dataset.
  - Tests with varying input sizes.
  - Tests on random, sorted, and reversely sorted inputs.
- Fill out the results tables and plot the results.
- Discuss the findings.

Once you have implemented the given algorithms in Java, you need to perform a set of experiments, report, illustrate, and analyze the results, and discuss your findings.

3.3.1 Experiments with Sorting Algorithms

In the given set of experiments, you will run the given sorting algorithms on different input types and sizes.

Experiments on the Given Random Data In the first set of experiments, you will test the algorithms on the given random datasets with varying input sizes. You are expected to measure the **average** running time of each algorithm **by running each experiment 10 times and taking the average of the recorded running times** for each input size. Make sure to save the sorted input data for the next set of experiments. Please be careful that you measure the running time of the sorting process only. To obtain the input numbers for each test case of size n, take the first n rows of the given dataset.

Report your findings in milliseconds (ms) by filling out the first five empty rows in Table 3.

Table 3: Results of the running time tests performed for varying input sizes (in ms).

	Input Size n									
Algorithm	**500**	**1000**	**2000**	**4000**	**8000**	**16000**	**32000**	**64000**	**128000**	**250000**
	Random Input Data Timing Results in ms									
Quick sort										
Insertion sort										
Merge sort										
Shell sort										
Radix sort										
	Sorted Input Data Timing Results in ms									
Quick sort										
Insertion sort										
Merge sort										
Shell sort										
Radix sort										
	Reversely Sorted Input Data Timing Results in ms									
Quick sort										
Insertion sort										
Merge sort										
Shell sort										
Radix sort										

Experiments on the Sorted Data In this second set of experiments, you should run your algorithms all over again, but this time on the already sorted input data that you obtained in the previous experiments. **The same averaging rule over 10 tries should also be applied in this step**. Fill out the next five empty rows in Table 3 with the new measured running times for the sorted input data.

Experiments on the Reversely Sorted Data In this third set of experiments, you should run your algorithms on the reversely sorted input data. You should use the already sorted input data that you obtained in the previous experiments and reverse them first (or simply read the sorted array from the end). Please make sure that **you measure the running time of the sorting process only**. **The same averaging rule over 10 tries should also be applied in this step**. Fill out the final five empty rows in Table 3 with the new measured running times for the reversely sorted input data.

3.3.2 Complexity Analysis and Result Plotting

After completing all the tests, you should analyze the obtained results in terms of the computational and auxiliary space complexity of the given algorithms. First, complete Tables 4 and 5, and justify your answers in short. Note that we are not interested in the overall space complexity of these algorithms, only in the additional memory space they use while performing the sorting operations. Please state which lines from the given pseudo-codes you used to obtain the auxiliary space complexity answers.

Table 4: Computational complexity comparison of the given algorithms.

Algorithm	**Best Case**	**Average Case**	**Worst Case**
Quick sort	$\Omega()$	$\Theta()$	$O()$
Insertion sort	$\Omega()$	$\Theta()$	$O()$
Merge sort	$\Omega()$	$\Theta()$	$O()$
Shell sort	$\Omega()$	$\Theta()$	$O()$
Radix sort	$\Omega()$	$\Theta()$	$O()$

Table 5: Auxiliary space complexity of the given algorithms.

Algorithm	**Auxiliary Space Complexity**
Quick sort	$O()$
Insertion sort	$O()$
Merge sort	$O()$
Shell sort	$O()$
Radix sort	$O()$

Plot the results obtained from the experiments in 3.3.1. **You should obtain separate plots for each sorting algorithm, showing performance across random, sorted, and reversely sorted data.** Table 3 should result in three plots, comparing all sorting algorithms with each-other under each input condition. For all plots, X-axis should represent the input size (the number of input instances n), while Y-axis should represent the running time in ms. See Figure 2 to see an example of how you should demonstrate your results.

The sample chart includes two different plotted algorithms, and illustrate how the performance

of a faster algorithm on average can degrade significantly in some worst case scenarios. Your plots must include the results of all given algorithms presented in a similar manner.

The plotting operation must be handled programmatically by using a readily-made *Java* libraries. You can leverage any library of your choice. You can also use the **XChart** library, which is open-source and pretty easy to use. To use the **XChart** library, you should first obtain the *.zip* file by using the download button from the following link https://knowm.org/open-source/xchart/. Then, you should extract the file and add **xchart-3.8.1.jar** to your project; i.e., include it in your classpath. You can check the example code provided by the authors from the link: https://knowm.org/open-source/xchart/xchart-example-code/.

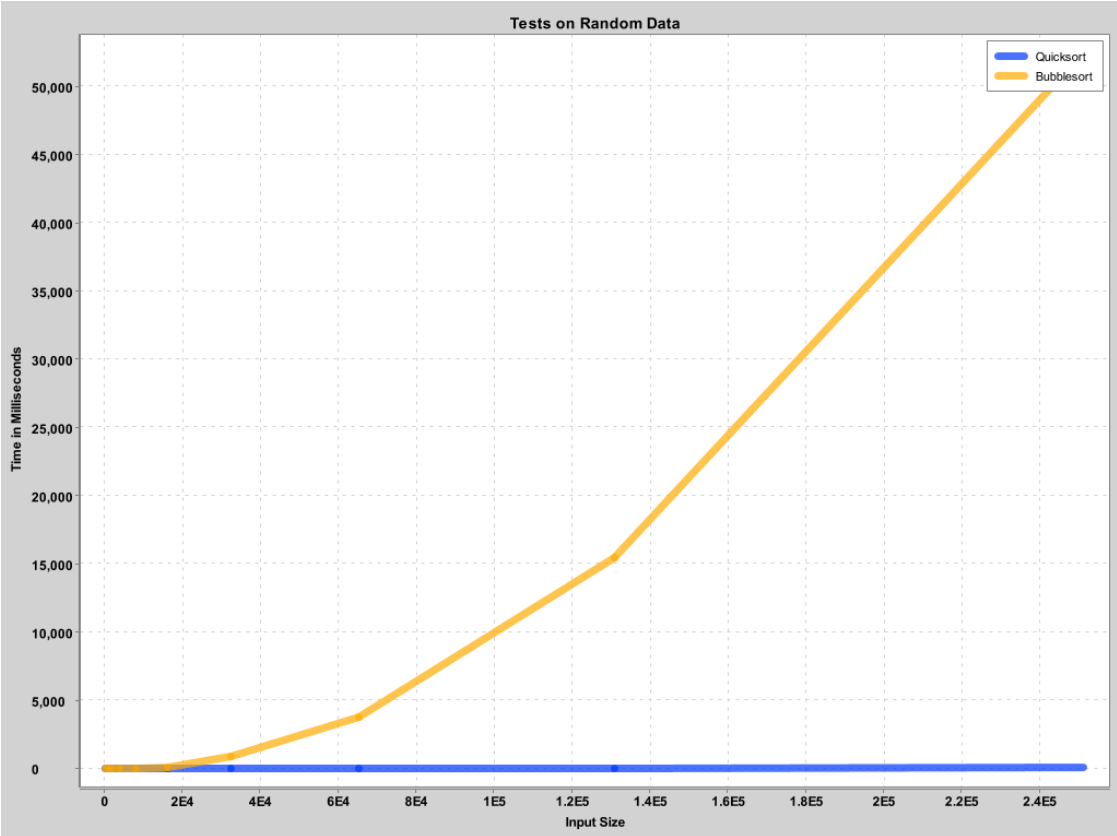


Figure 2: A sample plot showing running time results for varying input sizes on random input data for two sample sorting algorithms.

3.3.3 Results Analysis and Discussion

Briefly discuss the obtained results by referring to Table 4 and the obtained plots in 3.3.2. Answer the following questions:

- What are the best-, average-, and worst-case time complexities of the given algorithms? For each case, describe the input characteristics that trigger these behaviors.
- Do the empirical results (running times of your algorithm implementations) match the

theoretical asymptotic complexities? If there are discrepancies, explain the possible reasons.

- Compare the performances of Quick Sort and Merge Sort across different input orderings (random, sorted, reverse-sorted). Discuss how their algorithmic strategies influence performance consistency.
- Compare the performance of Shell Sort and Insertion Sort across different dataset sizes. At what input size does Shell Sort begin to significantly outperform Insertion Sort in your experiments, and why?
- Given that Radix Sort is a non-comparison-based sorting algorithm, discuss how it handles large numerical ranges.Under what conditions does Radix Sort outperform comparison-based algorithms?

Grading Policy

- Implementation of the algorithms: 20%
- Performing the experiments and reporting the results and making comments: 25%
- Completing the given two computational and auxiliary space complexity tables 15%
- Plotting the results: 20%
- Results analysis and discussion: 20%

What to Include in the Report

You are encouraged to use this **Programming Assignment Report Template** and create your reports in LaTeX. We suggest using Overleaf platform for this. This is not mandatory, but make sure your report has all necessary parts and information (click here to see the PDF example).

Your report needs to include the following:

1. Include a brief problem statement.
2. Include your Java codes corresponding to the given sorting algorithms.
3. Include all running time results tables corresponding to all given experiment sets performed on the given random, sorted, and reversely sorted data, for varying input sizes. All five algorithms must be tested.
4. Include two completed tables that show the theoretical computational and auxiliary space complexities of the given algorithms, with a brief justification of your answers.
5. Include at least four plots of the obtained results from step 3.
6. Briefly discuss the obtained results by answering the given questions.
7. In preparing the report, you may use any theoretical resources, online or otherwise, but make sure to include the references in your report. **Do not copy any ready codes from the internet as there is a big chance someone else could do the same**

thing, and you would get caught for cheating even if you don't know each other!

Important Notes

- Do not miss the deadline: **Friday, 20.03.2026 (23:59:59)**.
- Save all your work until the assignment is graded.
- The assignment solution you submit must be your original, individual work. Duplicate or similar assignments are both going to be considered as cheating.
- You can ask your questions via Piazza (https://piazza.com/hacettepe.edu.tr/spring2026/bbm204), and you are supposed to be aware of everything discussed on Piazza.
- You will submit your work via https://submit.cs.hacettepe.edu.tr/ with the file hierarchy given below:

```
b<studentID>.zip
├─ src.zip <FILE>
│  ├─ Main.java <FILE>
│  │
│  ├─ *.java <FILE>
│
├─ report.pdf <FILE>
```

- The name of the main class that contains the main method should be **Main.java**. You may use **this starter code** which has a helpful example of using **XChart** library. The main class and all other classes should be placed directly (no subfolders) into a zip file named **src.zip**.
- This file hierarchy must be zipped before submitted (not .rar, only .zip files are supported).

Academic Integrity Policy

All work on assignments **must be done individually**. You are encouraged to discuss the given assignments with your classmates, but these discussions should be carried out in an abstract way. That is, discussions related to a particular solution to a specific problem (either in actual code or in pseudocode) **will not be tolerated**. In short, turning in someone else's work (including work available on the internet), in whole or in part, as your own will be considered as **a violation of academic integrity**. Please note that the former condition also holds for online/AI sources. Make use of them responsibly, refrain from generating the code you are asked to implement. Remember that we also have have access to such tools, making it easier to detect such cases.

The submissions will be subjected to a similarity check. Any submissions that fail the similarity check will not be graded and will be reported to the ethics committee as a case of academic integrity violation, which may result in suspension of the involved students.

References

[1] "Sorting algorithm." Wikipedia, https://en.wikipedia.org/wiki/Sorting_algorithm, Last Accessed: 10/02/2022.

[2] N. Faujdar and S. P. Ghrera, "Analysis and Testing of Sorting Algorithms on a Standard Dataset," 2015 Fifth International Conference on Communication Systems and Network Technologies, 2015, pp. 962-967.

[3] G. Batista, "Big O," Towards Data Science, Nov 5. 2018, https://towardsdatascience.com/big-o-d13a8b1068c8, Last Accessed: 10/03/2024.